3-D Trajectory Design Based on Deep Reinforcement Learning for UAV-Assisted Communication Networks

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***Abstract*—Most of the existing UAV-assisted communication networks provide service only for static users or deterministically moving ones. In fact, for some complex and dynamically changing scenarios, the users communicating to the UAV may move randomly, with unpredictable mobility. The uncertainty of users' movements poses a challenge to the guarantee of stable network performance. To tackle this, the paper investigates a UAV-assisted communication network, where a UAV provides communication service for ground users which are moving randomly. We collectively factor in ground user mobility, task duration, and UAV flight restrictions to design precise 3D trajectory for UAV, and formulate them into an optimization problem, aiming to maximize the network throughput while minimizing UAV energy consumption. Considering the dynamics caused by users' uncertain movement, we transform the optimization problem into a Markov decision process (MDP), then improve the twin-delayed deep deterministic policy gradient (TD3) to design UAV's 3D trajectory. By utilizing the prior knowledge to accelerate the exploration efficiency, we propose a trajectory design algorithm based on prior knowledge-TD3 (PKTD3-TD), enabling UAV to autonomously adjust flight parameters by leveraging environmental observations under dynamic conditions for enhancing flexibility and intelligence. Simulation results show that our proposed scheme outperforms the compared ones in terms of communication link quality, network throughput and UAV's energy consumption.**

***Index Terms*—Wireless communication networks, UAV, 3-D trajectory design, deep reinforcement learning, prior knowledge.**

I. Introduction

In RECENT years, uncrewed aerial vehicle (UAV)-assisted communication has attracted significant attention, bringing forth numerous new possibilities and advantages [1], [2]. In such networks, UAV equipped with communication devices serves as aerial base station (BS), providing flexible deployment and extended coverage in areas with limited infrastructure [3], [4]. Additionally, UAV can act as relay node to support emergency response operations, assist in search and rescue missions, and even facilitate communication during large-scale events [5], [6]. In the context of the 6G communication network, UAV also plays a crucial role as they occupy the airspace, establishing a vital network layer between ground and space networks [7], [8], [9]. UAV can communicate with ground stations and satellite stations, forming a seamless space-air-ground network that lays the foundation for fully integrated 6G heterogeneous networks [10], [11].

Thanks to the high air maneuverability, UAV can approach users and establish high-probability line-of-sight (LoS) links, thereby providing better quality communication services [12], [13]. By optimizing UAV flight trajectories, these link advantages can be fully exploited. Furthermore, UAV trajectory optimization has a significant impact on key performance metrics, such as service time, network delay, energy consumption and power transmission [5], [14], [15], [16]. Among these, system throughput and UAV energy consumption are particularly critical, as they jointly determine the efficiency and sustainability of UAV-assisted communication systems. Therefore, optimizing the UAV flight trajectory to balance throughput and energy efficiency is essential for achieving practical and high-performance UAV-assisted networks.

Traditional optimization methods, such as successive convex approximation (SCA) [17], alternating optimization (AO) [18], and block coordinate descent (BCD) [19], have been widely applied to UAV trajectory design and resource allocation problems. However, these approaches often rely on idealized assumptions and the loss of precision caused by the non-convexity of optimization problems [20], [21]. In practical UAV-assisted communication systems, factors such as user mobility, channel variation, and flight constraints jointly affect communication performance, making it difficult for conventional mathematical optimization methods to achieve real-time adaptability. Fortunately, deep reinforcement learning (DRL) can capture nonlinear state–action mappings and enable real-time strategy adjustment based on environmental changes. DRL has demonstrated strong adaptability and effectiveness in complex and dynamic communication scenarios [22], [23], [24], [25].

Many studies have utilized DRL to learn and adapt to complexity and varying environments, designed and adjusted the flight trajectory of UAV to enhance system performance [26], [27], [28]. However, most DRL-based methods rely on simplified or semi-static settings, where users are stationary or channel variations are slow. As a result, UAV trajectories are designed under quasi-static assumptions with limited in-operation

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adjustments, restricting their applicability in realistic dynamic networks with user mobility and time-varying demands. In practice, UAV trajectory design is affected by multiple dynamic factors, including the surrounding environment, the served object, and the constraints of UAV itself. This requires the network model to be closer to the actual requirements and more influencing parameters to be taken into account. Moreover, to fully exploit UAV maneuverability, flight states such as altitude, speed, acceleration, direction, etc., should be flexibly adjusted to ensure that it can fly autonomously in 3-D space. In addition, the DRL algorithm used for UAV path planning should be able to deal with continuous states, and have fast convergence speed and efficient learning ability.

Motivated by these, we investigate 3-D trajectory design for UAV-assisted communication networks, where UAV has limited energy and serves mobile users over a large area within a finite time horizon. Considering practical factors including region constraints and UAV flight limitations, we formulate an optimization problem to enhance overall system performance. Due to the unpredictability of user mobility and continuous action space, we reformulate the optimization problem as a Markov decision process (MDP) and propose a trajectory design algorithm based on prior knowledge-the twin-delayed deep deterministic policy gradient (PKTD3-TD), which combines prior knowledge to effectively reduce the space and time of exploration. Under flight constraints and user mobility, the UAV can autonomously plan its trajectory to enhance system throughput while reducing energy consumption. The main contributions of this paper are summarized as follows.

- By comprehensively considering ground user mobility, task duration, UAV flight direction, speed, and acceleration, we jointly optimize the throughput and energy consumption of UAV-assisted communication network to design a 3-D dynamic trajectory for UAV in complex environments.
- Due to the stochastic nature of user mobility and continuous control of infinite motion space, traditional optimization methods struggle to solve this problem. Therefore, we reformulate the original problem as an equivalent MDP and propose a PKTD3-TD algorithm based on TD3, in which the UAV can autonomously select flight direction and speed for continuous movement according to environmental observations. Additionally, to accelerate convergence and improve stability of the algorithm, we introduce prior knowledge and design a prior knowledge policy to effectively explore the environment, and thereby enhancing the network performance.
- We compare our proposed scheme with other benchmark ones to evaluate its effectiveness. Simulation results validate that our proposed scheme can accurately extract users' mobility patterns, follow them, and establish robust communication links between UAV and moving users, thereby reducing unnecessary flights and achieving an improvement in terms of network throughput and UAV energy consumption compared with others.

The rest of this paper is organized as follows. Section II reviews the related works. In Section III, the UAV-assisted communication network model is described and the optimization problem is formulated. In Section IV, the PKTD3-TD algorithm is presented. Section V evaluates the performance of the proposed algorithm and Section VI summarizes the paper.

Notations: In this paper, scalars are denoted by italic letters, and vectors are denoted by boldface letters. The Euclidean norm of a vector is denoted by $\|\cdot\|$, and $\{\cdot\}$ denotes an array. The distribution of a Gaussian random noise with mean μ and covariance σ^2 is denoted by $\mathcal{N}(\mu, \sigma^2)$, $\mathbb{E}[\cdot]$ denotes the statistical expectation, $\bigtriangledown$ denotes the gradient of function, and $\mathrm{clip}(\cdot)$ denotes the boundary clipping function.

II. Related Work

In this work, we focus on optimizing the 3-D flight trajectory of UAV to enhance system throughput and reduce UAV energy consumption in dynamic user environments. To better clarify the contribution of this work, we provide a brief review of representative studies in the following section. A summary comparison between these studies and our proposed method is presented in Table I.

A. UAV-Assisted Communication Networks

1) Static UAV-Assisted Communication Networks: In static UAV-assisted communication networks, ground users are assumed to remain fixed, allowing UAV trajectory design and resource allocation to be optimized under stable channel conditions. For example, reinforcement learning (RL)-based approaches were explored in [29], [30] to enhance data collection efficiency and reduce energy consumption under constrained or imperfect CSI conditions. To address complex 3-D flight dynamics and environmental constraints, [31] and [32] designed efficient and feasible 3-D UAV trajectories based on an improved ant colony optimization algorithm and an optimal control framework, respectively. In addition, the works in [33] and [34] jointly optimized UAV trajectory, resource allocation, and device scheduling to improve system performance. Meanwhile, studies such as [35], [36], [37] further incorporated user clustering and visiting sequence design, aiming to enhance throughput while reducing mission completion time or energy consumption. Additionally, the research focus has progressively expanded from single-UAV systems to more complex network models. Multi-UAV and NOMA-enabled frameworks in [38], [39] were proposed to improve energy efficiency and fairness, while blockage-aware and edge-computing-based designs in [40], [41] focused on reducing latency and improving computational efficiency. Furthermore, physical-layer security enhancement [42] and joint trajectory and infrastructure deployment [43] have also been considered to extend system functionality.

Existing studies on static UAV-assisted communication networks have achieved remarkable progress in trajectory design, resource allocation, and mission scheduling under fixed user positions. Nevertheless, most of them are developed under the assumption of stationary ground users and stable channel conditions, which largely simplifies the problem but fails to capture the dynamic and time-varying characteristics of practical scenarios.

2) Dynamic UAV-Assisted Communication Networks: In dynamic UAV-assisted communication networks, user mobility and time-varying channels introduce significant challenges for

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TABLE I
COMPARISON WITH EXISTING SCHEMES

Reference	3-D trajectory	Mobility of users	Total throughput	UAV energy consumption	Wide range of activity	Prior knowledge	Reinforcement learning	Deep neural networks
[30]			✓				✓	
[44]		✓		✓			✓	✓
[45]		✓	✓				✓	✓
[46]		✓			✓	✓	✓	✓
[47]	✓	✓	✓		✓		✓	✓
[48]	✓	✓	✓		✓		✓	✓
[72]			✓	✓			✓	✓
[74]	✓		✓	✓	✓		✓	✓
[77]		✓		✓	✓		✓	✓
[78]				✓				
[79]				✓	✓			
[99]					✓		✓	✓
[100]				✓			✓	✓
[101]	✓			✓	✓		✓	✓
[102]						✓	✓	✓
[109]	✓				✓	✓	✓	✓
[112]		✓		✓	✓	✓	✓	✓
[113]	✓			✓	✓	✓	✓	✓
Our work	✓	✓	✓	✓	✓	✓	✓	✓

trajectory planning and resource allocation. For example, the authors of [44] employed BCD and SCA methods to jointly optimize trajectory, power, and base station selection, achieving a balance between sensing and communication performance. In [45], an online trajectory and resource allocation strategy based on Lyapunov optimization was developed to balance system energy efficiency in scenarios with user mobility. [46] formulated a joint task offloading and trajectory optimization problem to enhance secure computing under user mobility. [47] integrated digital twin and generative models to propose a coalition game–based resource collaboration mechanism, where virtually pre-generated coalition structures reduced scheduling complexity and improved energy efficiency. The works in [48], [49], [50] applied PPO, DQL, and dueling DQL algorithms to optimize UAV trajectory and communication strategies, aiming to balance throughput, energy consumption, and service delay. To further enhance learning performance, prior knowledge and rule-guided exploration were incorporated in [51], while multi-agent DRL frameworks were developed in [52], [53] to improve scalability, throughput, and fairness in large-scale systems.

Research on dynamic UAV-assisted communication networks has increasingly focused on trajectory design under user mobility and time-varying channel conditions, with most works limited to 2D path optimization. Existing methods often fail to fully capture the spatial dynamics and environmental uncertainty of practical UAV operations, leaving room for further exploration of more realistic and adaptive trajectory optimization frameworks.

B. Optimization Objectives for UAV-Assisted Communication Networks

1) Single-Objective Optimization: In single-objective formulations, the UAV trajectory is optimized to maximize or minimize a specific metric, such as throughput, delay, or energy consumption. For energy-oriented design, [54] jointly optimized hovering position and visiting order using forward–backward and AO methods, while [55] developed a fine-grained energy model to co-optimize flight frequency and trajectory for minimizing total energy consumption. In addition, [56], [57] focused on improving energy efficiency in space–air–ground integrated networks via joint power and trajectory design. Delay and information-oriented objectives have also been widely studied. A Transformer-based trajectory planning scheme combined with weighted A$*$ was proposed in [58] to reduce the Age of Information (AoI), while [59], [60] designed trajectory and scheduling strategies to minimize task completion time and average delay. Furthermore, [61] considered 3-D multi-trajectory planning under physical constraints to reduce delivery time. In the context of mobile edge computing (MEC), [62], [63] jointly optimized UAV trajectory, computation offloading, and resource allocation to reduce user energy consumption and ensure system stability. Additional studies have addressed more complex system considerations. Work in [64] addressed radar-assisted missions by jointly optimizing UAV 3D trajectory, scheduling, and power allocation through a three-layer iterative algorithm. Further, [65], [66], [67] applied iterative optimization frameworks combining BCD and SCA to enhance throughput and user experience. To ensure safe flight, [68] imposed continuous trajectory constraints for no-fly zones, reducing communication interruptions.

Single-objective optimization has deepened the understanding of UAV trajectory design, but its focus on a single metric limits overall system performance. A more comprehensive multi-objective framework is required to balance conflicting goals such as throughput, delay, and energy efficiency.

2) Multi-Objective Optimization: Multi-objective optimization jointly considers metrics such as throughput, coverage, and energy efficiency to achieve balanced system performance. In heuristic-based approaches, [69], [70], [71], [72] employed ant colony optimization, improved two-stage heuristics, genetic algorithms with constraint-repair, and artificial hummingbird optimization, respectively, to jointly optimize UAV trajectory, hovering position, and resource allocation, balancing energy consumption, task completion time, and coverage. Optimization-theoretic methods have also been widely studied. The works in [73], [74], [75] formulated joint optimization

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problems involving task offloading, resource allocation, and UAV control, solved via Lyapunov optimization and iterative frameworks to balance delay, energy consumption, and quality of experience (QoE). More recently, DRL-based methods have been adopted for complex trade-off problems. For instance, [76], [77] jointly optimized trajectory, power allocation, and communication performance utilizing the TD3 algorithm and multi-agent reinforcement learning, respectively, achieving an adaptive balance among data rate, AoI, and energy consumption. For instance, [78], [79] combined DRL with heuristic algorithms and beamforming techniques to further improve data collection efficiency, reduce delay, and enhance communication security.

Recent studies have achieved significant progress in jointly optimizing key performance metrics such as throughput, energy efficiency, and latency, effectively improving overall network balance. However, most existing works remain scenario-specific and computationally intensive, with limited adaptability to highly dynamic UAV-assisted communication environments.

C. Optimization Methods for UAV-Assisted Communication Networks

1) Model-Based Optimization Methods: Model-based approaches rely on accurate mathematical modeling of the communication system, where optimization problems are typically formulated as convex, mixed-integer, or non-convex programs. [80], [81], [82] respectively adopted mathematical programming methods and iterative algorithms combining SCA and BCD to minimize the total service time and AoI. [83] proposed a Rapidly-exploring Random Tree path planning method combining greedy strategies for efficient path planning. Regarding system-level joint optimization, [84], [85], [86] utilized methods such as Lyapunov optimization and geometric programming to jointly optimize UAV trajectory, communication scheduling, power allocation, task offloading and caching, aiming to enhance system throughput and energy efficiency. For more complex non-convex formulations, a variety of iterative optimization techniques have been employed. For example, [87], [88] utilized alternating optimization and SCA methods to handle imperfect CSI, multi-interference, and secure communication design, while [89], [90] combined SCA with semi-definite relaxation to jointly optimize hybrid beamforming, power allocation, and UAV trajectory for improved throughput and secrecy performance. In addition, [91], [92], [93] developed advanced iterative frameworks integrating AO, SCA, and convex transformation techniques to efficiently solve joint optimization problems involving trajectory control, resource allocation, and power management. A distributed approach in [94] decomposed user association and UAV deployment via coalition formation and particle swarm optimization to maximize user QoE.

Despite the progress of model-based optimization, such methods rely on accurate mathematical modeling and iterative solutions. Their complexity grows rapidly with the system scale increases. Moreover, they usually adopt static or quasi-static assumptions. In dynamic environments with mobile users, they require frequent re-optimization and cannot meet real-time requirements. This motivates the adoption of DRL with stronger real-time decision-making capability.

2) DRL-Based Optimization Methods: With the growing complexity and dynamics of UAV-assisted communication networks, DRL has become a powerful alternative to traditional model-based optimization methods. Unlike conventional approaches that require repeated problem reformulation and iterative solving, DRL learns a direct mapping from states to actions through interaction with the environment, enabling efficient online decision-making in high-dimensional and dynamic scenarios. Typical algorithms such as DQN [95], DDPG [96], PPO [97], and TD3 [98] have been widely applied to UAV trajectory planning, resource allocation, and power control. For instance, [30] proposed an attention-based DRL scheme to reduce energy consumption while maintaining data quality, while [99] developed a TD3-based method to minimize data collection time under imperfect CSI. Similarly, [100] introduced a Q-learning framework to improve data value with reduced energy consumption. To address complex system coupling, multi-agent DRL frameworks have been extensively explored. The works in [101], [102] adopted multi-agent DDPG to jointly optimize UAV trajectory and resource allocation, effectively handling user mobility and time-varying tasks, while [103] proposed a multi-agent PPO algorithm for ISAC systems to jointly optimize communication and sensing performance. Moreover, [104] integrated empirical channel knowledge into DNN-based path planning, while [105] designed a multi-objective 3D trajectory algorithm optimizing user access rate, fairness, and throughput. [106] integrated meta-learning and computation-awareness into DRL to improve throughput while reducing energy consumption and latency. A backward PPO algorithm with a feedback mechanism was presented in [107] to enhance sampling efficiency and mapping accuracy. Furthermore, [108] developed an online PPO-based method enabling UAV to autonomously adjust flight attitude, minimizing mission completion time.

Although DRL-based approaches significantly improve adaptability and decision-making capability, most existing studies focus primarily on policy optimization, with limited exploration of incorporating prior knowledge to guide learning and accelerate convergence.

D. Application of Prior Knowledge in Learning-Based Optimization

Recent studies have explored the incorporation of prior knowledge into learning-based optimization across various domains. For example, in UAV communications, [109] constructed an environmental-channel knowledge map integrating building blockage and channel gain priors into 3D trajectory planning to enhance transmission quality. [110] utilized environmental and task priors to assist multi-UAV energy-efficient trajectory planning, while [111] employed an outage-probability knowledge map to guide trajectory optimization in cellular networks. Prior knowledge also plays a vital role in other scenarios. [112] introduced explicit environmental priors and label-continuity

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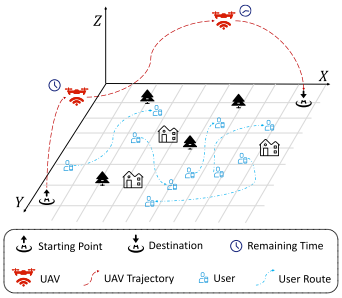


Fig. 1. Network model.

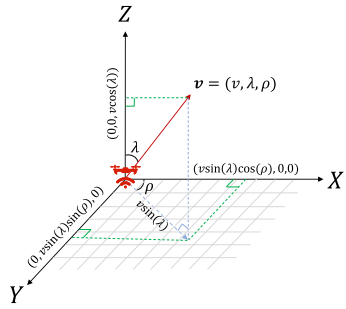


Fig. 2. The spherical coordinate.

constraints to enhance cross-domain generalization for physiological signal estimation, realizing robust cross-modality transfer. In streaming data, [113] presented a transferable prior shift framework to dynamically embed external knowledge into Bayesian models, alleviating concept drift and prior forgetting. For 3D medical image multitask learning, [114] designed a dual-prior scheme to guide feature decoupling via task difficulty and anatomical structure priors.

These studies indicate a clear trend toward dynamically adaptive and task-aware use of prior knowledge, enhancing the learning efficiency and stability of data-driven models. For UAV-assisted communication networks, where the state space is continuous and the environment is time-varying, effectively leveraging prior knowledge to accelerate convergence and reduce redundant exploration remains an open challenge.

III. Network Model and Problem Formulation

A. Network Model

As shown in Fig. 1, we consider a UAV-assisted communication network, where a UAV with limited energy acts as an aerial base station to collect data from mobile ground users, such as audio and video, traffic flow information, etc. It is assumed that both the UAV and ground users are equipped with single omni-directional antennas. The UAV communicates with users by employing frequency division multiple access (FDMA). K ground users are randomly distributed and move freely. The positions of the UAV and the k-th user at time t are denoted by $\boldsymbol{q}_t = [x_t, y_t, z_t] \in \mathbb{R}^3$ and $\boldsymbol{w}_t^k = [x_t^k, y_t^k, 0] \in \mathbb{R}^3$, $0 \le t \le T$ respectively, where $[x_t, y_t]$ denotes the projection of the UAV's position onto the horizontal plane at time t, z_t stands for the UAV's altitude, and $[x_t^k, y_t^k]$ denotes the horizontal coordinates of the k-th user, T denotes the maximum flight time of the UAV. To facilitate the optimization of the UAV's trajectory, we discretize the continuous time domain into N time slots with the duration δ, $\delta = T/N$.

B. UAV Mobility Model

In UAV-assisted communication networks, the UAV needs to utilize various states observed in the current time slot to plan its path in the next time slot. When making decisions, the positions of the UAV and the mobile users in the next time slot are required. Therefore, we model the UAV's mobility and the users' movement respectively to predict their positions in the next time slot.

As shown in Fig. 2, we utilize spherical coordinates $\boldsymbol{v} = (v, \lambda, \rho)$ to model the UAV's mobility, where $v = \|\boldsymbol{v}\|, v \in [0, v_{\max}]$ denotes the UAV's flight velocity, $v_{\max}$, λ, and ρ are the maximum flight velocity, the polar angle, and the azimuth angle on the horizontal plane, respectively, $\lambda \in [0, \pi]$, $\rho \in [-\pi, \pi]$. Note that in this paper, the direction of velocity $\boldsymbol{v}$ aligns with the direction of acceleration $\boldsymbol{ac}$. Assuming that the UAV's flight velocity, polar angle and azimuth angle in the current time slot n, $n \in \{1, \ldots, N\}$, its positions in the $(n+1)$-th time slot are predicted as

$$x_{n+1} = x_n + v_n \sin(\lambda_n \cos(\rho_n)\delta, \forall n \in \{1, N\} \quad (1)$$

$$y_{n+1} = y_n + v_n \sin(\lambda_n) \sin(\rho_n)\delta, \forall n \in \{1, N\} \quad (2)$$

$$z_{n+1} = z_n + v_n \cos(\lambda_n)\delta, \forall n \in \{1, N\} \quad (3)$$

where x_n, y_n, and z_n are UAV's observed 3-D coordinates in the current time slot, x_{n+1}, y_{n+1}, and z_{n+1} are the predicted position coordinates at the next time slot. It is noted that when the UAV or mobile users detect their position or posture data, the observed values will replace the predicted ones.

C. User Mobility Model

In this paper, the UAV serves the ground users, and the users can move randomly over a large area. The UAV gets the positions of the mobile users in current time slot through communication, but it can't know them in the next time slot. When designing the trajectory, the UAV needs to estimate the users' positions in the next time slot. To characterize the movement of the users and estimate their locations in the next time slot, like [115], a Gaussian-Markov mobility model is adopted to predict the users' locations in the next time slot.

It is supposed that the UAV obtains the positions, moving velocities and directions of all users through communication in the n-th time slot. The velocity v_{n+1}^k and direction ϑ_{n+1}^k of user k in the $(n+1)$-th time slot are predicted as

$$v_{n+1}^k = \omega v_n^k + (1-\omega)\overline{v}_n + \sqrt{1-\omega^2}\psi_k, \forall n \in \{1, N\} \quad (4)$$

$$\vartheta_{n+1}^k = \omega \vartheta_n^k + (1-\omega)\overline{\vartheta}_n + \sqrt{1-\omega^2}\varphi_k, \forall n \in \{1, N\} \quad (5)$$

where v_n^k and ϑ_n^k are the velocity and direction of user k in the $(n+1)$-th time slot respectively. $\overline{v}$ and $\overline{\vartheta}$ are the average velocity and direction of all users. ω is a tuning parameter

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used to control the level of randomness, $0 \leq \omega \leq 1$. ψ_k and φ_k are two independent Gaussian distributions for $\mathcal{N}(0, \sigma_1)$ and $\mathcal{N}(0, \sigma_2)$ respectively, reflecting the randomness of user k in the movement patterns. Based on formula (4) and (5), the positions of user k in the $(n+1)$-th time slot are predicted as

$$x_{n+1}^k = x_n^k + v_{n+1}^k \cos(\vartheta_{n+1}^k)\delta, \forall n \in \{1, N\} \tag{6}$$

$$y_{n+1}^k = y_n^k + v_{n+1}^k \sin(\vartheta_{n+1}^k)\delta, \forall n \in \{1, N\} \tag{7}$$

where x_n^k and y_n^k are the observed position coordinates in the n-th time slot, x_{n+1}^k and y_{n+1}^k are the predicted position coordinates in the $(n+1)$-th time slot.

D. Channel Model

Since this work deals with the 3-D dynamic trajectory planning of UAV in urban environment and the users are moving, the type of propagation environment and the elevation angle significantly influence the air-to-ground channel. Similar to [116], [117], we adopt the probabilistic path loss model for LoS links. In the n-th time slot, the LoS probability between the UAV and user k can be expressed as

$$P_{n,\mathrm{LoS}}^k = \frac{1}{1 + b_1 \exp\left(-b_2 \left(\frac{180}{\pi} \arcsin\left(\frac{z_n}{r_n^k}\right)\right) - b_1\right)} \tag{8}$$

where b_1 and b_2 are constants that depend on the type of environment, and $r_n^k = \|q_n - w_n^k\|$ is the distance between the UAV and the user k in the n-th time slot. Therefore, the probability of a non-line-of-sight (NLoS) link can be given by

$$P_{n,\mathrm{NLoS}}^k = 1 - P_{n,\mathrm{LoS}}^k \tag{9}$$

Continuing, the path loss for LoS and NLoS links between the UAV and user k in the n-th time slot can be expressed as

$$PL_{n,\mathrm{LoS}}^k = LF_n^k + \eta_{\mathrm{LoS}} \tag{10}$$

$$PL_{n,\mathrm{NLoS}}^k = LF_n^k + \eta_{\mathrm{NLoS}} \tag{11}$$

where $LF_n^k = 20\log(r_n^k) + 20\log(f_c) + 20\log(4\pi/v_c)$ denotes the free-space path loss between the UAV and user k, f_c is the carrier frequency and v_c is the speed of light, η_{LoS} and η_{NLoS} denote the propagation loss of LoS and NLoS links, respectively. The overall path loss between the UAV and user k can be expressed as

$$PL_n^k = P_{n,\mathrm{LoS}}^k \times PL_{n,\mathrm{LoS}}^k + P_{n,\mathrm{NLoS}}^k \times PL_{n,\mathrm{NLoS}}^k. \tag{12}$$

Then, the transmission rate R_n^k between the UAV and user k in the n-th time slot can be expressed as

$$R_n^k = \frac{B_u}{K} \log_2\left(\frac{p_k}{\varrho_n^k N_0 B_u / K}\right) \tag{13}$$

where B_u is the total bandwidth of the UAV, p_k is the transmission power of user k, N_0 is the power spectral density of the noise channel, and $\varrho_n^k = 10^{PL_n^k/10}$[116].

The total transmission rate of the UAV in the n-th time slot can be expressed as

$$R_n = \sum_{k=1}^{K} R_n^k. \tag{14}$$

E. UAV's Energy Consumption Model

In UAV-assisted communication networks, UAV is usually powered by onboard batteries, so its energy is limited and it is sensitive to the energy consumption. The energy consumption of the UAV includes two main components, namely the communication-related energy and propulsion energy. The communication-related energy is used in communication circuits of the UAV to send, receive and process the signals, which is negligible compared to the propulsion energy [118]. Therefore, we only consider the propulsion energy consumption of the UAV. Similar to [119], the propulsion power consumption of the UAV is

$$\begin{aligned} P(\tau_c, F_y, v_n) = {} & P_0\left(1 + \frac{3v_n^2}{U_{tip}^2}\right) \\ & + (1+m)\frac{(W - F_y/\cos\tau_c)^{3/2}}{\sqrt{2\rho A}} \\ & \times \left(\sqrt{\tilde{m}^2 + \frac{v_n^4}{4{V_0}^4}} - \frac{v_n^2}{2{V_0}^2}\right)^{1/2} \\ & + \left(W - F_y\right) v_n \tan\tau_c + \frac{1}{2}\rho s A v_n^3 d_0 \cos^2\tau_c \end{aligned} \tag{15}$$

where $\tau_c = \pi/2 - \lambda$ and $F_y = \frac{1}{2}C\rho S_{FP} V_y^2$ denote the UAV's pitch angle and longitudinal air resistance, respectively. Here, C, S_{FP}, and V_y are the air resistance coefficient, fuselage equivalent flat plate area, and longitudinal flight speed. v_n is the flight speed of UAV in the n-th time slot, where $v_n = v_{n-1} + \boldsymbol{ac}_n \cdot \delta$, and $\boldsymbol{ac}_n$ is the acceleration of the UAV in the n-th time slot. P_0 is the blade profile power of the UAV in hovering state and ρ is the air density. m is the induced power correction factor, while $\tilde{m} = \frac{W - F_y}{W\cos\tau_c}$ characterizes the effect of the UAV's flight attitude and aerodynamic drag on the induced velocity. W, s, and A represent the UAV's weight, rotor solidity, and rotor disc area. d_0 is the drag ratio of the fuselage and V_0 is the mean rotor induced velocity in hovering. Therefore, the UAV's energy consumption in the n-th time slot can be expressed as

$$E_n = P(\tau_c, F_y, v_n) \cdot \delta. \tag{16}$$

F. Problem Formulation

In practical applications, the energy consumption of UAV has a great impact on trajectory planning. Our goal is that the network achieves maximum throughput with minimum energy consumption of UAV. Since the maximum throughput and the minimum energy consumption are opposite to each other, to maintain the consistency of the two optimization objectives, we reverse the energy consumption and make it $-E_n$. In this way, minimizing energy consumption becomes maximizing $-E_n$. Therefore, we define a composite metric named the difference of throughput and energy (DTE) to optimize the network performance. Moreover, to adapt to various scenarios, where some networks with abundant energy focus more on improving communication performance while other energy-constrained systems are more sensitive to energy consumption, we introduce weight

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coefficients into the optimization objectives to adjust the proportion of importance for different objectives according to actual needs. For instance, in extreme cases, if an energy-rich network has no concerns about energy but only focuses on improving communication performance, the weight of energy consumption in its objective function can be set to zero. Conversely, in a network with limited energy, the weight of energy consumption should be larger so that the system can complete the required tasks as much as possible, while the weight of throughput can be set to be relatively small or even zero. Therefore, the objective function can be expressed as DTE, which evaluates the weighted difference between the total throughput and the energy consumption throughout the task duration. That is

$$DTE = \sum_{n=1}^{N} (w_1 R_n - w_2 E_n) \tag{17}$$

where w_1 and w_2 are the weighted coefficients of throughput and energy consumption respectively, $w_1 \geq 0$, $w_2 \geq 0$.

It is worth noting that although there are significant differences in terms of physical meaningfulness and order of magnitude between throughput and energy consumption, their physical units are set to bits^{-1} and Joules^{-1} respectively, adjusting w_1 and w_2 can align their numerical scales and maintain the consistency in form. This treatment ensures the additivity and numerical consistency of two objectives with distinct physical properties in a unified optimization goal, thereby transforming the DTE into a dimensionless scalar reward. Meanwhile, by choosing appropriate w_1 and w_2, the network can trade-off the throughput and energy consumption based on actual needs and optimize the network performances, thereby enhancing system applicability. Furthermore, since the throughput and energy consumption are mutually coupled and influence each other in UAV-assisted communication networks, DTE merely unifies the two contradictory objectives in a formal sense. To simplify the analysis, some assumptions are made. Firstly, the system targets long-term cumulative utility, where step-wise throughput and energy consumption can be calculated independently, and their instantaneous coupling is neglected. Secondly, it is assumed that the power consumed by UAV remains constant within a time slot.

By optimizing UAV's flight direction, speed, and acceleration/deceleration, the trajectory planning problem can be formulated as

$$\begin{aligned}
\mathbf{P}: &\max_{\{x_n, y_n, z_n\}, N} DTE \\
s.t.\ C1: &\ N \leq \frac{T}{\delta} \\
C2: &-\pi \leq \rho_n \leq \pi, \forall n \\
C3: &\ 0 \leq \lambda_n \leq \pi, \forall n \\
C4: &\ 0 \leq v_n \leq v_{\max}, \forall n \\
C5: &\ 0 \leq \|\boldsymbol{ac}_n\| \leq ac_{\max}, \forall n \\
C6: &\ \boldsymbol{q}_0 = \boldsymbol{q}_s, \boldsymbol{q}_N = \boldsymbol{q}_e \\
C7: &\ x_{\min} \leq x_n \leq x_{\max}, \forall n \\
C8: &\ y_{\min} \leq y_n \leq y_{\max}, \forall n \\
C9: &\ z_{\min} \leq z_n \leq z_{\max}, \forall n
\end{aligned} \tag{18}$$

where $ac_{\max}$ is the maximum acceleration of the UAV. $\boldsymbol{q}_s$ and $\boldsymbol{q}_e$ are the UAV's starting and destination points respectively. $x_{\min}$, $x_{\max}$, $y_{\min}$ and $y_{\max}$ are the limits on the xy-plane of the flying area, $z_{\min}$ and $z_{\max}$ are the constraints on the UAV's flight height. The UAV must travel from the starting point $\boldsymbol{q}_s$ to the destination $\boldsymbol{q}_e$ within a specified time frame. $C1$ enforces that the UAV's flight duration does not exceed the maximum time. $C2$-$C5$ impose restrictions on the UAV's actions during flight. $C6$ ensures that the UAV must depart from the starting point and reach the destination. $C7$-$C9$ define the feasible flight space for the UAV.

Since the optimization problem above is non-convex and belongs to the class of NP-hard problems, resembling the traveling salesman problem. As the number of users and the dimensionality of the system state increase, conventional optimization-based formulations typically require frequent problem reformulation and iterative solving, which significantly limits their scalability in dynamic scenarios. Meanwhile, the continuous mobility of ground users leads to time-varying system states, making the traditional methods difficult to adapt to environmental changes in a timely manner. In contrast, DRL provides a model-free framework that learns a direct mapping from system states to continuous UAV control actions through environment interaction, without relying on explicit mobility prediction. This enables adaptive and robust decision-making in dynamic environments while avoiding repeated optimization. Based on the above considerations, in the next section, we propose a DRL-based solution to the optimization problem.

IV. Trajectory Design Based on DRL

In this section, we propose a 3-D trajectory design algorithm named PKTD3-TD to maximize the weighted difference of network throughput and UAV flight energy consumption. The UAV serves as an intelligent agent, exploring the environment to gather experiences and optimize its flight strategy. We have meticulously designed the state space, action space and reward function. Additionally, we incorporate prior knowledge to accelerate algorithm convergence and enhance stability.

A. MDP Design

RL is a machine learning technique with the goal of learning excellent control strategies. The agent learns its policy through trial and error by interacting with the environment [120]. In this work, the UAV trajectory design problem is formulated as a MDP. Both the state and action spaces are defined as continuous to accommodate the UAV's control requirements for flight direction, velocity, and acceleration. In the n-th time slot, the agent obtains the current state information s_n from the environment, makes a decision action a_n based on the current state information to reach the next state s'_{n+1} and gets a reward. This process is repeated until an excellent policy is achieved. Therefore, we first design the key components of the MDP: state, action and reward.

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1) State s_n in the n-th time slot: The state information in DRL denotes the environment information perceived by the agent and the changes brought about by its actions [121]. The design of states is crucial for convergence and performance of DRL algorithms. To characterize the fundamental coupling between UAV trajectory control and communication performance, the MDP state is expressed in a noise-free form as a baseline representation. Below, we provide a detailed design of the state.

a) *UAV's location information:* $\boldsymbol{q}_n$. We have defined a flight area $C7$-$C9$ for the UAV. The agent must be aware of its own position to avoid flying out of bounds. Therefore, the state space should include the UAV's position.

b) *User's location information:* $[x_n^k, y_n^k], k \in K$. We design trajectory for the UAV with the objective of maximizing throughput while minimizing UAV energy consumption. As transmission rate (13) highlights, the transmission rate between the UAV and users is significantly influenced by the magnitude of path loss. The value of path loss is determined based on the positions of both the UAV and users. Therefore, we include the users' location information $[x_n^k, y_n^k]$ in the state representation.

c) *UAV's speed:* v_n. As constrained by $C5$, we set an acceleration boundary, when making the next flight decision, the agent must consider the current velocity to avoid violating acceleration constraints. Therefore, the state space should include the current flight velocity in the current time slot.

d) *Remaining time and distance:* $[t_n^{re}, d_n^{re}]$. $t_n^{re} = T - n\delta$ denotes the remaining time, and $d_n^{re} = \|q_e - q_n\|$ denotes the distance between the UAV and the destination. As indicated by constraints $C1$ and $C6$, the UAV must reach the destination within the specified time. The agent should be aware of its remaining mission time and the distance to the destination in order to make decisions: whether to continue serving users or reach the destination in a timely manner.

Finally, the state s_n is defined as

$$s_n = \left\{x_n, y_n, z_n; x_n^1, y_n^1, \ldots, x_n^K, y_n^K; v_n; t_n^{re}, d_n^{re}\right\}. \quad (19)$$

The agent obtains all the information it needs in state s_n, which corresponds to the optimization problem $\mathbf{P}$ we aim to solve. The state s_n has a dimension of $2K + 6$.

2) Action a_n in the n-th time slot: The agent's actions correspond to the UAV's flight control under constraints $C2$-$C4$, the UAV's flight actions include flight direction and flight velocity. Specifically, the action a_n is composed of the flight speed $v_n \in [0, v_{\max}]$, the polar angle $\lambda_n \in [0, \pi]$, and the azimuth angle $\rho_n \in [-\pi, \pi]$ selected at the n-th time slot. Therefore, the action chosen by the agent in state s_n is

$$a_n = \{v_n, \lambda_n, \rho_n\} \quad (20)$$

3) Reward r_n in the n-th time slot: The design of reward determines whether the agent can learn the desired policy. Our optimization objective is to maximize the weighted difference of network throughput and UAV flight energy consumption. In each time slot, there is throughput generated between the UAV and users, this is precisely the main event we want the agent to explore, corresponding to the mainline reward, we set this as a positive reward. It's important to note that only rewards related to throughput are positive. The throughput portion of the reward is defined as

$$r_{n,1} = c_{th} R_n \delta \quad (21)$$

where c_{th} is a positive constant used to scale down the throughput, as excessively large rewards can lead to the 'gradient vanishing' problem in the Tanh$(\cdot)$ activation function.

In each slot, the UAV's flight energy consumption is one of the main events we want the agent to explore, and we hope the UAV to consume as little energy as possible. Therefore, the energy consumption part of the reward is

$$r_{n,2} = -c_{en} E_n \delta \quad (22)$$

where c_{en} is a positive constant used to scale down the energy consumption. Next is the reward setting for constraint $C6$. The reward for the agent to reach the destination promptly before time runs out is written as

$$r_{n,3} = \begin{cases} 0, & N < \frac{T}{\delta} \\ -((1 - \xi_n^{ar}) \cdot c_{ar} \cdot \frac{d_N^{re}}{v_{\max}} + c_{nr}), & N = \frac{T}{\delta} \end{cases} \quad (23)$$

where ξ_n^{ar} is a binary arrival indicator, $\xi_n^{ar} = 1$ denotes the agent reaching the destination, and 0 otherwise, c_{ar} is the penalty coefficient for exceeding the time limit, and c_{nr} is an additional penalty term for not reaching the destination. Notice that for the event of reaching the destination, we do not provide any positive reward. On the contrary, if the agent fails to reach the destination within the specified time T, a significant penalty is imposed. The only positive reward the agent can receive is related to throughput, this design encourages the agent to reach the destination on time and maximize throughput. Triggering the event of reaching the destination is sparse in our task, the agent is unlikely to explore this event, and thus the reward $r_{n,3}$ obtained by the agent at the end of each exploration episode is almost always negative. This can make the algorithm struggle to converge. To address this issue, we introduce an additional reward component for the agent, which is

$$r_{n,4} = -t_n^{\text{lack}} \left(1 - c_{\text{near}} \frac{d_n^{\text{near}}}{v_{\max}}\right) \quad (24)$$

where t_n^{lack} denotes the time deficit of the UAV, d_n^{near} denotes the decrease in distance from the UAV's current position to the destination compared to the previous time slot, and $0 \le c_{\text{near}} \le 1$ is a scaling coefficient. t_n^{lack} and d_n^{near} are

$$t_n^{\text{lack}} = \min\left(\max\left(\frac{d_n^{re}}{v_{\max}}, 0\right), c_{\text{lack}}\right) \quad (25)$$

$$d_n^{\text{near}} = \|\boldsymbol{q}_{n-1} - \boldsymbol{q}_s\| - \|\boldsymbol{q}_n - \boldsymbol{q}_s\| \quad (26)$$

where c_{lack} is a positive constant used to bound the penalty. Based on the three equations mentioned above, we can observe that when the remaining time t_n^{re} becomes smaller compared to t_n^{re}, the value of t_n^{lack} becomes larger, leading to a higher penalty value for $r_{n,4}$. The UAV can reduce this penalty value by getting closer to the destination. Conversely, moving away from the destination increases this penalty value. The coefficient $r_{n,4}$ ensures that this penalty remains significant when time is insufficient. Next is the reward settings for agents violating flight restrictions. In the previous action design, we have already ensured the speed

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constraint during UAV flight. Additionally, we need to consider acceleration constraint $C5$ and flight area constraints $C7$-$C9$, these two parts of the reward are expressed as

$$r_{n,5} = -c_{ac} \cdot \xi_n^{ac} \tag{27}$$

$$r_{n,6} = -c_h \cdot \xi_n^{h} \tag{28}$$

where ξ_n^{ac} denotes a binary acceleration constraint indicator, $\xi_n^{ac} = 1$ indicates that the agent violates the acceleration constraint, and 0 otherwise. Similarly, ξ_n^h is a binary height constraint indicator. c_{ac} and c_h are positive penalty constants associated with violating constraints $C5$ and $C9$, respectively. Note that we do not impose penalties for $C7$ and $C8$ because our designated destination is located on the xy-plane boundary. Due to the nature of reaching the destination, the UAV agent will unavoidably violate constraints $C7$ and $C8$ during its trial-and-error process. When the agent violates the xy-plane constraint without reaching the destination, we cancel the action taken by the agent.

The reward obtained by the agent for selecting action a_n given the state s_n is represented as

$$r_n = r_{n,1} + r_{n,2} + r_{n,3} + r_{n,4} + r_{n,5} + r_{n,6}. \tag{29}$$

B. PKTD3-TD Algorithm

To enhance the UAV trajectory optimization process, we propose a prior knowledge-guided TD3 algorithm named PKTD3-TD. Since the TD3 algorithm may face challenges such as slow convergence and excessive negative rewards in UAV-assisted communication networks [122], [123], we introduce prior knowledge to guide the UAV's initial exploration, enabling the agent to accumulate valuable experience and accelerate convergence. The prior knowledge is developed into the TD3 framework and integrated with three key mechanisms, including clipped double Q-learning, delayed policy updates, and target policy smoothing regularization, to enhance the stability and convergence efficiency of policy optimization. Distinctively, this framework establishes a reactive closed-loop control policy, where action decisions rely solely on the current system state, enabling real-time responses to environmental changes.

1) Design prior knowledge for the proposed algorithm: To maximize the weighted difference of throughput and energy consumption, the agent must consider not only mainline events related to throughput and energy consumption but also destination arrival events. Although the reward function is carefully designed, purely random exploration in the early training stage often generates low-quality experiences due to energy penalties and delayed rewards. When such experiences dominate the replay buffer, they can significantly slow down convergence and destabilize the training process. To address this issue, prior knowledge is introduced to guide exploration. By incorporating action-level guidance, the agent can better understand the task and environment, thereby improving learning efficiency. Below, we will outline the steps involved in introducing prior knowledge.

The agent's initial policy is to fully explore the target area within the specified time. We specify the flight direction as any direction within the given plane, while the velocity is randomly chosen. The actions based on prior knowledge are defined as

$$a_n^{pk} = \{\mathrm{rand}(0,1) \cdot v_{\max}, \lambda_{pk}, \mathrm{rand}(0,1) \cdot \rho_{pk}\} \tag{30}$$

where $\mathrm{rand}(0,1)$ denotes a random number within (0,1), λ_{pk} and ρ_{pk} denote the polar angle and the maximum adjustable azimuthal angle, respectively, for UAV flight as specified by the prior knowledge. In cases with distinct starting and destination points, this setting prevents the UAV from prematurely taking the shortest path, forcing it to first explore the area and gather rich throughput experiences. Conversely, for identical starting and destination points, it prevents the UAV from being trapped at the origin. In the initial stage of training, we employ the method of prior knowledge to gradually add successful experiences to the experience replay buffer. Specifically, define the maximum number of experiences using prior knowledge as R_{rand}. During the initial R_{rand} steps, the agent doesn't rely on the policy network's output for actions. Instead, it performs the prior actions from (30) to populate the experience replay buffer. In the subsequent training steps, the agent's actions are generated by the policy network. So, the way the agent selects an action is expressed as

$$a_n = \begin{cases} a_n^{pk}, & R_{ex} \le R_{rand} \\ \mathrm{clip}(\pi_\phi(s_n) + \epsilon, -c, c), & R_{ex} > R_{rand} \end{cases} \tag{31}$$

where R_{ex} is the number of experiences in the experience replay buffer, and $\epsilon \in \mathcal{N}(0, \sigma_3)$ is Gaussian noise. By integrating such prior knowledge with random exploration organically, the UAV is capable of rapidly discovering new optimal flight directions when facing sudden changes in user movement, thereby facilitating rapid adaptation to the network.

2) Clipped double Q-Learning: $Q(s,a) = \mathbb{E}[G_n|s,a]$ represents the action-value function, indicating the expected return when the agent takes action a in state s. TD3 employs two critic networks and uses the minimum of their target Q-values to compute the target, which helps mitigate overestimation bias associated with a single Q-value estimate. The target update in the clipped double Q-learning algorithm is

$$y = r + \gamma \min_{i=1,2} Q_{\theta_i'}(s', a'), a' \sim \pi_{\phi'}(s') \tag{32}$$

where y is the target value, r is a reward, $Q_{\theta_i'}(i = 1, 2)$ are two target critic networks with parameters θ_i', $\pi_{\phi'}$ is the target actor network with parameters ϕ', and a' is the action selected by the target actor network based on the next state s'.

3) Target network and delayed policy updates: Unlike DDPG [124], where actor and critic networks are updated synchronously. In TD3, the actor network and three target networks are updated less frequently compared to the critic network. As a result, the target policy network lags behind the current policy, further reducing potential overestimation and fluctuations. The update mechanisms for the actor network and critic network are represented as

$$\bigtriangledown_\phi J(\phi) = M_b^{-1} \sum \bigtriangledown_a Q_{\theta_1}(s,a)|_{a=\pi_\phi(s)} \bigtriangledown_\phi \pi_\phi(s) \tag{33}$$

$$\theta_i = \arg\min_{\theta_i} M_b^{-1} \sum (y - Q_{\theta_i}(s,a))^2, i = 1, 2 \tag{34}$$

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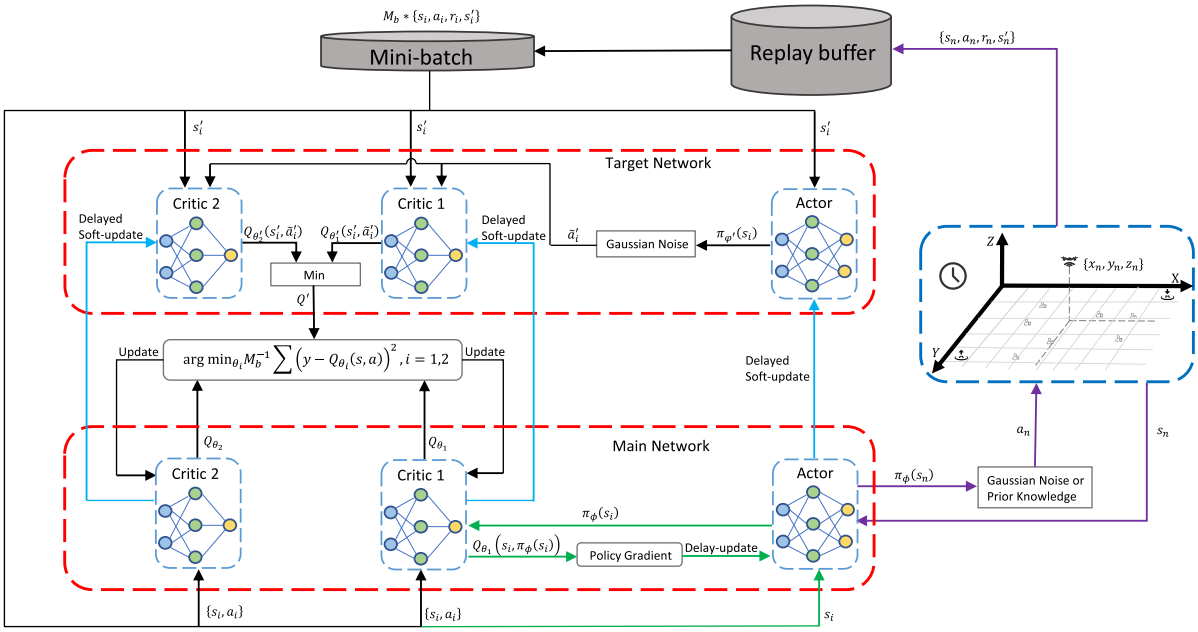


Fig. 3. PKTD3-TD network framework.

where M_b is the size of the mini-batch, $\pi_\phi(\cdot)$ refers to the policy network (i.e. actor network) parameterized by ϕ, and $Q_{\theta_i}(i = 1, 2)$ are two critic networks with parameters θ_i. The update of the target network is

$$\theta' = \tau\theta + (1-\tau)\theta' \tag{35}$$

where θ' is the target network parameter, θ is the parameter of the target network's corresponding main network, and τ is the soft update coefficient.

4) Target policy smoothing regularization: DDPG tends to overestimate and suffer from function approximation errors, thereby increasing the variance of target values. TD3 employs regularization to mitigate this induced variance. Specifically, when computing the target value, random noise is added to the actions output by the target actor network, effectively smoothing the action space and reducing errors from overestimation. The target value y is calculated as

$$y = r + \gamma \min_{i=1,2} Q_{\theta_i'}(s', \widetilde{a}) \tag{36}$$

$$\widetilde{a} = \mathrm{clip}(\pi_{\phi'}(s') + \epsilon, -c, c) \tag{37}$$

$$\epsilon \in \mathrm{clip}\left(\mathcal{N}(0, \widetilde{\sigma}), -\frac{c}{2}, \frac{c}{2}\right). \tag{38}$$

where c is the action boundary value.

Specifically, the combination of clipped double Q-learning, delayed policy updates, and target policy smoothing regularization suppresses value overestimation and policy fluctuations, thereby ensuring stable decision outputs in dynamic environments. Simultaneously, the integration of prior knowledge and the exploration mechanism with random noise enables the agent to incorporate fresh transitions and adjust its flight policy rapidly in response to mobility mutations. Furthermore, it is worth noting that PKTD3-TD optimizes long-term cumulative rewards rather than instantaneous state matching, which encourages the policy to focus on overall communication performance trends instead of reacting over-sensitively to transient state variations. As a result, the learned policy remains stable even when state observations are inaccurate or delayed.

C. Flow of the PKTD3-TD Algorithm

In Fig. 3, we present the PKTD3-TD network framework for UAV-assisted communication systems. By integrating prior knowledge with the TD3 algorithm, the proposed method optimizes UAV trajectories to maximize the weighted difference between throughput and energy consumption. The detailed pseudo-code is provided in Algorithm 1.

The PKTD3-TD algorithm begins by initializing the critic and actor networks with parameters θ_1, θ_2, and ϕ (*Line* 1), copying them to the corresponding target networks (*Line* 2), and initializing the experience replay buffer R (*Line* 3). During interaction, the environment is reset at the beginning of each episode, and the initial state s_0 is obtained (*Line* 5). Based on the size of the experience replay buffer, the agent generates an action using the prior policy or adds noise to the action from the policy network $\pi_\phi(\cdot)$ and executes it (*Line* 7-8), ensuring both guided exploration and sufficient randomness. The environment checks if the action violates the altitude and acceleration constraints and takes appropriate actions (*Lines* 9-14). Then, the reward and next state are updated (*Line* 15), and the experience tuple is stored in the experience replay buffer R for network updates (*Line* 16). After sufficient samples are collected (*Line* 17), mini-batches are drawn from R to update the critic networks using target policy smoothing (*Lines* 18–21). This mechanism allows the algorithm to reduce the risk of over-reliance on a single strategy. Every d time slots (*Line* 22), the actor network is updated using the deterministic policy gradient (*Lines* 23-24), and the target networks are updated (*Lines* 25-27).

It is noticed that in practical UAV-assisted communication systems, the proposed PKTD3-TD algorithm is generally

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Algorithm 1: PKTD3-TD.

Initialize critic networks $Q_{\theta_1}, Q_{\theta_2}$ and actor network π_ϕ with random parameters θ_1, θ_2, and ϕ
Initialize target networks $\theta_1' \leftarrow \theta_1, \theta_2' \leftarrow \theta_2$ and $\phi' \leftarrow \phi$
Initialize experience replay buffer R
for episode=0 to M **do**
  Set the current time $n = 0$, place the UAV at the starting point $\boldsymbol{q}_s$ with velocity $v = 0$, randomly generate the user's position, $\overline{v}$ and $\overline{\vartheta}$, and obtain the initial state s_0
  repeat
    Select and execute an action a_n:
    $$a_n = \begin{cases} a_n^{pk}, & R_{ex} \le R_{rand} \\ \text{clip}(\pi_\phi(s_n) + \epsilon, -c, c), & R_{ex} > R_{rand}. \end{cases}$$
    if the agent exceeds the spatial constraints, i.e., $\xi_n^{xy} = 1$ or $\xi_n^h = 1$ **then**
      Cancel this action, i.e., the agent's position remains unchanged
    end if
    if the agent exceeds the acceleration constraints, i.e., $\xi_n^{ac} = 1$ **then**
      $v_{n+1} = v_n + \text{clip}(\|\boldsymbol{ac}_n\|, -ac_{\max}, ac_{\max})$
    end if
    Calculate the reward $r_n = r_{n,1} + r_{n,2} + r_{n,3} + r_{n,4} + r_{n,5}$ and obtain the next state s'_{n+1}
    Store the transition $\{s_n, a_n, r_n, s'_{n+1}\}$ in R
    if $R_{ex} > R_{rand}$ **then**
      Sample mini-batch of M_b transitions $\{s, a, r, s'\}$ from R
      Set $y = r + \gamma \min_{i=1,2} Q_{\theta_i'}(s', \widetilde{a})$
      Update the critic network by minimizing the loss:
      $\arg\min_{\theta_i} M_b^{-1} \sum (y - Q_{\theta_i}(s,a))^2, i = 1, 2$
      if n mod d **then**
        Update the actor network by the deterministic policy gradient:
        $M_b^{-1} \sum \bigtriangledown_a Q_{\theta_1}(s,a)|_{a=\pi_\phi(s)} \bigtriangledown_\phi \pi_\phi(s)$
        Update target networks:
        $\theta_i' = \tau\theta_i + (1-\tau)\theta_i', i = 1, 2$
        $\phi' = \tau\phi + (1-\tau)\phi'$
      end if
    end if
    $n \leftarrow n + 1$
  until $n = N_{\max}$ or $\xi_n^{ar} = 1$
end for

implemented in two phases: the offline training phase and the online execution phase. In the offline training phase, as can be seen from Algorithm 1, the critic networks are updated at every time slot, while the actor and target networks are updated every d time slots. The setting of time slot width depends on UAV's hardware configuration and the computational complexity of the algorithm. This phase involves updating one actor and two critic networks, and the computational complexity is $O(3 \times \sum_{i=1}^{L} d_i \cdot d_{i-1})$, where d_i represents the number of neurons in the i-th layer and L is the total number of hidden layers. When the time slot width is set to be greater than the execution time of the hardware, the algorithm can work in the actual system. Since the offline training phase is computationally intensive, it is usually conducted on a ground-based server or an edge computing platform. For the online execution phase, only the trained actor network is deployed onboard the UAV to take control actions based on the observed system states through forward inference, so the update time is one time slot. The minimum time slot width is determined by the execution time of the algorithm performed by the UAV, and the execution time depends on UAV's hardware and the computational complexity of the algorithm during its online execution phase. The higher the hardware configuration is, the smaller the minimum time slot width can be set and the shorter the update time will be. Since the online execution phase does not involve experience replay or network parameter updates, the computational complexity is reduced to $O(\sum_{i=1}^{L} d_i \cdot d_{i-1})$ for a forward inference of the actor network. This significantly leads to lower computational overhead and is suitable for UAV platforms with limited onboard computational resources.

By incorporating prior knowledge, the proposed PKTD3-TD algorithm enables faster convergence by guiding the agent toward higher-quality decisions. However, prior knowledge may be imperfect or incomplete. To address this, the algorithm leverages three mechanisms. Firstly, employing the exploration mechanism of DRL, the PKTD3-TD algorithm constantly interacts with the real environments and adapts its trajectory based on real-time system information, thereby reducing reliance on inaccurate prior knowledge. Secondly, the introduction of experience replay buffers and random noise expands the diversity of search strategies, and weakens the influence of inaccurate or incomplete prior knowledge. By updating experience replay buffers and smooth target policy in the training process, the PKTD3-TD algorithm can replace imperfect prior knowledge with effective experiences, thus overcoming the defects of inaccurate or incomplete prior knowledge. Finally, a multi-dimensional reward function, combining performance objectives and constraint penalties, guides the agent to weigh different objectives and search for an optimum in a wider strategy space when prior knowledge is insufficient or inaccurate.

V. Simulation Results

A. Simulation Setup

We evaluate our proposals in Python 3.8 and PyTorch 2.0 on a standard desktop workstation equipped with an Intel i7 CPU. The wireless communication environment is rigorously constructed using the standard scientific computing library NumPy based on the system models defined in Section III. We further validate the network performance using standard wireless channel models in MATLAB.

A service area of 600 m $\times$ 600 m is considered, the minimum flight altitude $z_{\min}$ for UAV is set to 50 m, and the maximum altitude $z_{\max}$ is 200 m. The starting point of the UAV is $\boldsymbol{q}_s = [0, 0, 50]$, and the destination is $\boldsymbol{q}_e = [600, 600, 50]$. In the above simulation environment, the average online execution time of the trained actor network for a decision-making was

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TABLE II
REWARD SETTING

Parameters	Values	Parameters	Values
c_{th}	$1/(2\times10^7)$	c_h	0.5
c_{ar}	1	c_{en}	$1/300$
c_{nr}	20	c_{near}	0.5
c_{ac}	0.5	c_{lack}	1

TABLE III
ALGORITHM PARAMETER SETTING

Parameters	Values	Parameters	Values
M	6000	C	0.2
R	10^5	W	20
M_b	128	S_{FP}	0.0151
γ	0.96	P	80.182
τ	0.005	A	0.503
c	1	s	0.05
d	2	ρ	1.23
λ_{pk}	0.5π	$\boldsymbol{d}_0$	1.23
ρ_{pk}	0.5π	U_{tip}	120
R_{rand}	20000	$\sigma_1, \sigma_2, \sigma_3, \widetilde{\sigma}$	$1, 0.25\pi, 0.1, 0.2$

measured to be approximately 0.1 ms, so the time slot is fixed as $\delta = 1$s, and the maximum flight time T is equal to 200 s. During online execution, the UAV updates its trajectory once per time slot by performing a single forward inference of the trained actor network. The initial speed of the UAV is $v_0 = 0$ m/s, and the maximum flight speed is $v_{\max} = 20$ m/s [99], and the maximum acceleration $ac_{\max} = 5$ m/s^2 [118]. We assume that the transmit power of each user $p_k = 10$ dBm, and the UAV's total bandwidth $B_u = 20$ MHz [99]. Additionally, the channel-related parameters are $b_1 = 9.61$, $b_2 = 0.16$, $\eta_{\text{LoS}} = 1$ and $\eta_{\text{NLoS}} = 20$ [117]. The weight coefficients of network throughput and energy consumption are $w_1 = 1/(2\times10^7)$, and $w_2 = 1/300$. The settings related to reward are summarized in Table II. These parameters are configured based on multi-objective optimization and numerical stability considerations, to balance different reward components and enforce task and flight constraints, thereby ensuring stable training and convergence. For Algorithm 1, the learning rate is set to 0.0001 [99]. All the actor and critic networks are constructed using a 2-layer fully connected feedforward neural network with 256 neurons. The actor network then applies the tanh$(\cdot)$ function as the activation to constrain the actions, while the critic network uses the ReLU$(\cdot)$ function for activation, and both actor and critic networks use the Adam optimizer. Table III summarizes the parameters for the UAV energy consumption model [118] and other settings [124].

In order to evaluate the performance of the proposals, we compare the proposed PKTD3-TD scheme with four baseline schemes as follows.

- TDPK: Trajectory design based on prior knowledge (TDPK) employs prior knowledge to guide that the UAV flies directly to the destination, and the flight speed for each time slot is randomly generated.
- Dueling DQL [47]: This method balances UAV's flight time and achieves throughput, allowing adjustment of the scaling factor to shorten the UAV's trajectory and appropriately increase data collected from users.
- PPO [44]: The PPO algorithm is used for UAV 3D trajectory planning with the same state and reward settings as the proposed method. At each time slot, the agent generates continuous control actions to update the UAV's flight direction and speed based on the learned policy.
- Greedy algorithm [83]: The greedy algorithm makes decisions based on the current system performance. At each time slot, it selects the action that maximizes the immediate objective function to pursue a local optimal solution under the current system state.

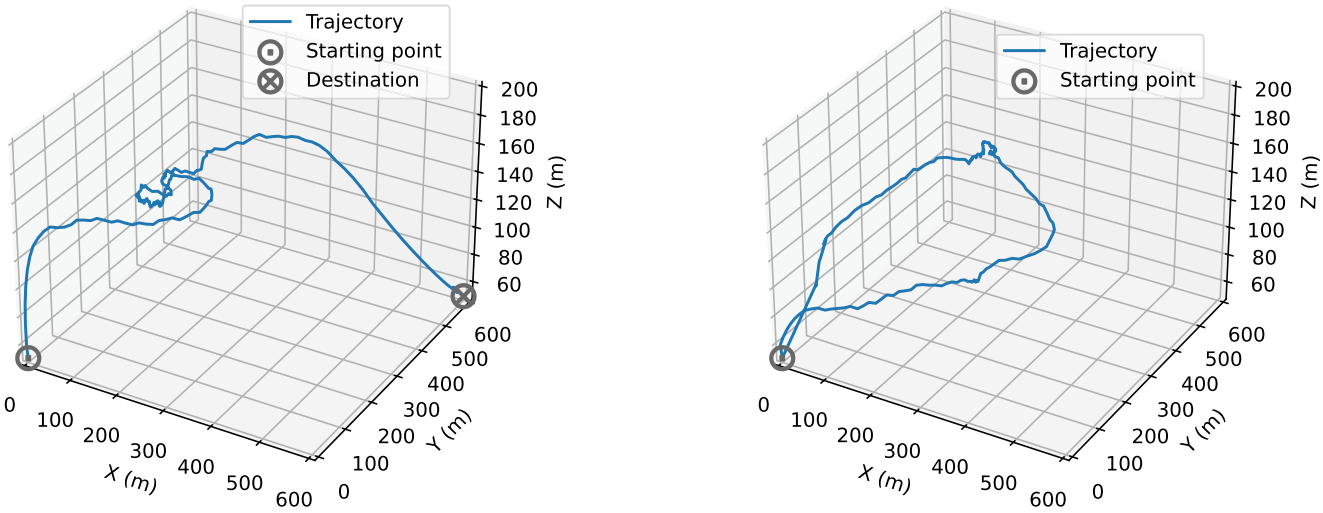
B. Results and Analysis

To verify the effectiveness of the proposed scheme, we conduct tests by using the trained model. Fig. 4 displays the 3-D trajectory of the UAV with the same and different starting and destination points. From Fig. 4, whether the starting and destination are the same or different, we can observe that the UAV quickly ascends to a higher flight altitude after taking off from the starting point. Subsequently, it adjusts its position based on the changes of user positions in a timely manner. As the time limit approaches, the UAV will fly towards its destination and ultimately reach it with precision. From Fig. 4, we can see that our approach applies to various configurations for starting and destination points. In the following discussion, we will mainly focus on the case where the starting and ending points are different.

Fig. 5 exhibits the projection of UAV and user locations onto the horizontal plane at time slots n =35, 70, 105, 140, 175, and 194 when the starting and ending points are different, and the number of users on which UAV services is 10. When sufficient time slots are available (e.g., n = 35-140), the UAV learns to align its horizontal position with the user cluster center, while maintaining a relatively high altitude. Starting from an initial altitude of 50 m, the UAV rapidly ascends (e.g., reaching 140.1 m at $n = 35$) to improve communication conditions when far from users, and then adaptively adjusts its trajectory toward user-dense regions to enhance data collection. When time is limited (e.g., n = 175), the UAV gradually distances itself from users and moves towards the destination. The corresponding altitude first increases and then decreases. This is because the learned policy by the UAV optimizes the long-term cumulative reward, where the importance of reaching the destination increases as the remaining time decreases, enabling an adaptive trade-off between communication performance and task completion.

In Fig. 6, we present the real-time average LoS probability and transmission rate during the UAV's flight on both the Python-based simulation platform and the MATLAB-based network evaluation platform. Specifically, using the UAV trajectories generated by the PKTD3-TD algorithm implemented in Python, the corresponding network performance is evaluated in Matlab with the same channel model and system parameters. Fig. 6(a) illustrates the real-time variation of the average LoS probability on both platforms. It can be observed that within the first 25 time slots, the LoS probability increases significantly, as the learned policy encourages the UAV to increase its flight altitude when moving away from users. Between $n = 25$ and $n = 170$,

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(a) UAV's trajectory with different starting and destination points.

(b) UAV's trajectory with the same starting and destination points.

Fig. 4. UAV's 3-D trajectory in different cases.

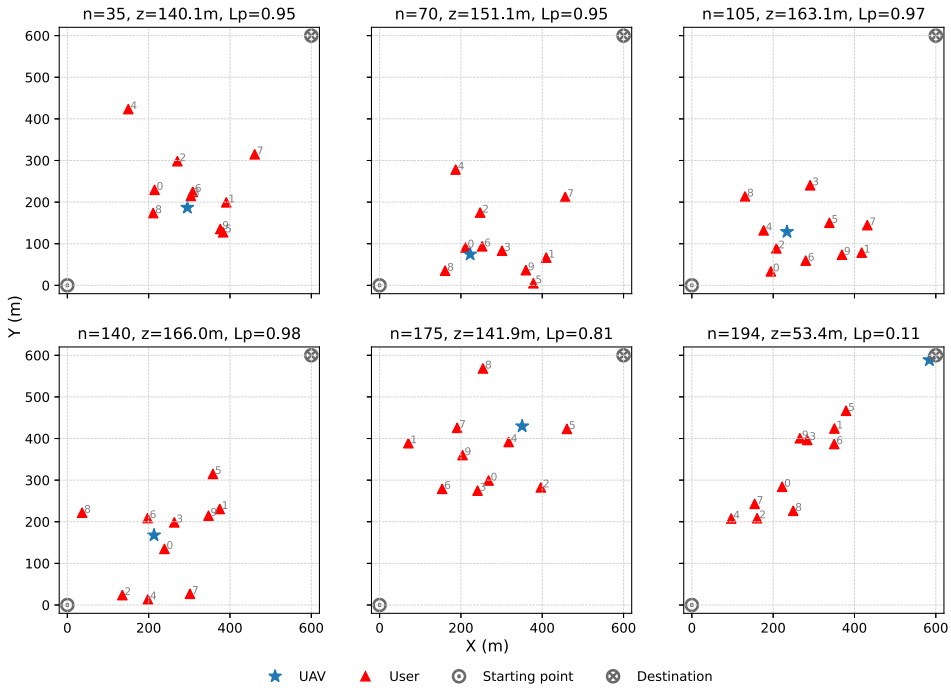
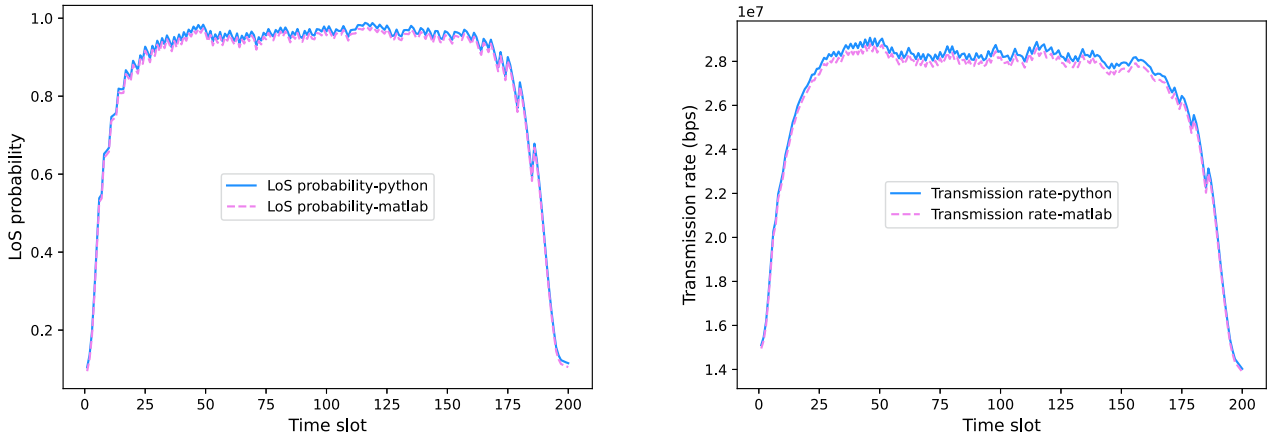


Fig. 5. Locations of UAV and users at different time slots (n: Time slot, z: UAV's flight altitude, Lp: LoS probability).

the UAV maintains a relatively stable altitude (as shown in Figs. 4 and 5), thereby sustaining a high LoS probability. After $n = 170$, due to the requirement of reaching the destination within a limited time, the UAV gradually moves away from users, leading to a decrease in LoS probability. Fig. 6(b) presents the corresponding transmission rate on both platforms. The transmission rate follows a similar trend to the LoS probability, with a rapid increase in the initial stage, a relatively stable performance in the middle stage, and a decline in the final stage. These results demonstrate that the PKTD3-TD algorithm adaptively learns how to adjust the flight trajectory based on real-time user positions and remaining time, thereby improving transmission performance. In addition, the results show that the real-time average LoS probability and transmission rate obtained from Matlab closely match those from the Python implementation across all flight processes. The highly consistent trend validates the accuracy of the communication modeling in the Python-based DRL environment and further validates the reliability of the proposed PKTD3-TD algorithm under realistic network performance evaluation.

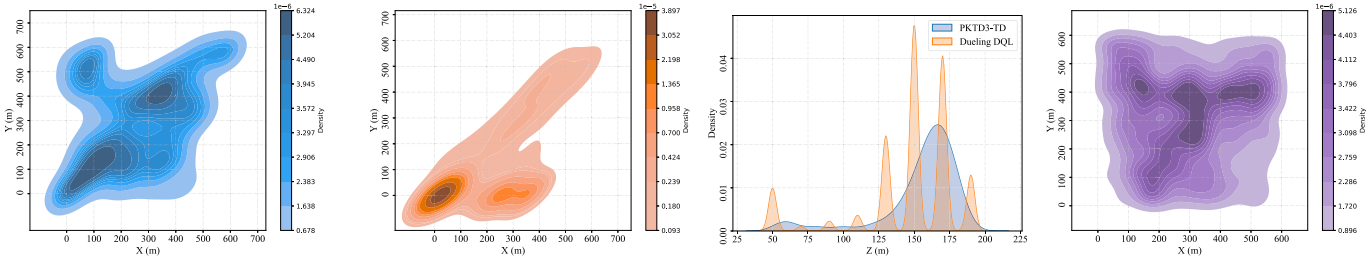
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(a) Real-time LoS probability.

(b) Real-time transmission rate.

Fig. 6. Comparison of real-time LoS probability and transmission rate obtained from Python-based simulation and Matlab-based network evaluation during UAV flight.



(a) The kernel density distribution of UAV position in the xy plane with the PKTD3-TD algorithm.

(b) The kernel density distribution of UAV position in the xy-plane with Dueling DQL.

(c) The kernel density distribution of UAV flight altitude with PKTD3-TD and Dueling DQL algorithms.

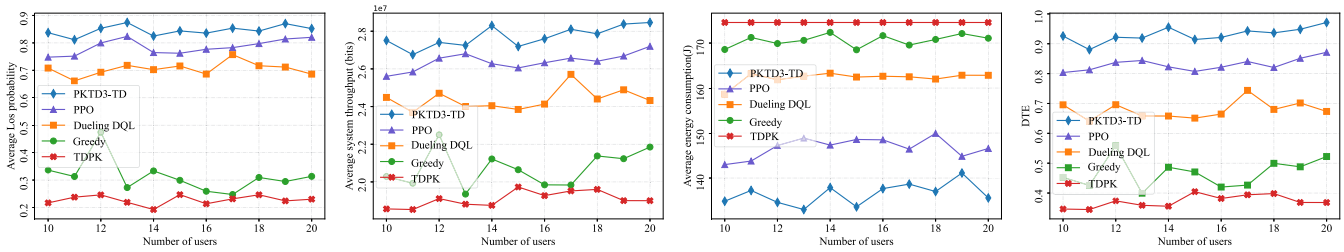
(d) The kernel density distribution of user positions in the x-y plane.

Fig. 7. UAV's position distribution in 10 flights with different methods and the user's position distribution.

To further validate the effectiveness of the proposed algorithm, we conduct 10 flights using different methods and record UAV positions. Fig. 7(a) and (b) depict the kernel density maps of the horizontal flight trajectories for PKTD3-TD and Dueling DQL algorithms, where darker regions indicate higher visit frequency. The PKTD3-TD algorithm generates a wider and more dispersed flight area, while Dueling DQL shows a more confined distribution. Fig. 7(c) illustrates the kernel density plots of flight altitude for the two algorithms. Both methods maintain relatively high altitudes to ensure LoS connectivity; however, Dueling DQL exhibits discrete altitude levels due to its discrete action space. This limits its ability to select optimal flight directions and altitudes, leading to more frequent adjustments and higher energy consumption. Fig. 7(d) shows the kernel density of mobile users distribution. Comparing with Fig. 7(a)–(b), the PKTD3-TD trajectories align more closely with user locations, indicating better adaptability to user mobility and adjust UAV flight paths in real-time to enhance communication link quality. Overall, the results confirm the superior adaptability of the PKTD3-TD algorithm in dynamic environments with mobile users.

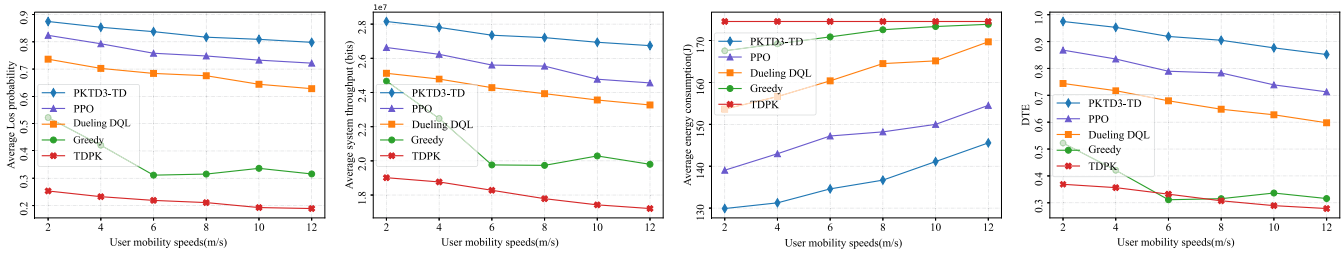
Fig. 8 displays network performances of compared schemes versus different user numbers. In Fig. 8(a) and (b), the proposed PKTD3-TD algorithm consistently outperforms all benchmarks. PPO benefits from continuous control but is less effective to track user distributions, while Dueling DQL fails to capture user mobility, resulting in limited trajectory diversity and lower throughput. The greedy method suffers from short-sighted decisions and severe LoS degradation, leading to far inferior to the RL-based schemes, while TDPK method ensures reaching the destination, it fails to fully utilize the UAV's 3-D spatial mobility, resulting in lower LoS probability and network throughput. In contrast, PKTD3-TD enables flexible 3-D trajectory optimization within the mission duration, thereby achieving superior throughput. This observation highlights the scalability of the proposed PKTD3-TD algorithm in multi-user scenarios. The reasons are as follows. According to (13) and (14), given a bandwidth, the system throughput is determined by the transmission rate, which is affected by the channel quality. The channel quality may be reflected by LoS probability. In Fig. 8(c), we compare the average energy consumption of different methods versus varying numbers of users. The PKTD3-TD algorithm achieves the lowest

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(a) Average LoS probability. (b) Average throughput. (c) Average energy consumption. (d) Average DTE.

Fig. 8. Average LoS probability, throughput, energy consumption and DTE versus different number of users. (a) Average LoS probability. (b) Average throughput. (c) Average energy consumption. (d) Average DTE.



(a) Average LoS probability. (b) Average throughput. (c) Average energy consumption. (d) Average DTE.

Fig. 9. Average LoS probability, throughput, energy consumption and DTE versus different user mobility speeds. (a) Average LoS probability. (b) Average throughput. (c) Average energy consumption. (d) Average DTE.

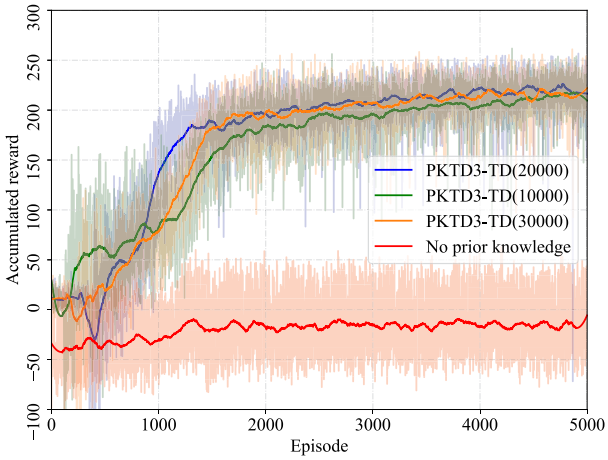
average energy consumption. With 15 users, it reduces energy consumption by 14.97 J, 30.843 J, 34.899 J and 42.972 J compared to PPO, Dueling DQL, greedy, and TDPK, respectively. This is attributed to reduced redundant movements and more efficient trajectory planning, whereas Dueling DQL and greedy algorithm suffer from inefficient adjustments due to discrete actions, and PPO incurs additional overhead due to unguided exploration. Fig. 8(d) shows the average weighted difference of throughput and energy consumption versus different numbers of users. With 15 users, the PKTD3-TD algorithm improves the weighted metric by 0.107, 0.263, 0.443, and 0.510 compared to PPO, Dueling DQL, greedy, and TDPK, respectively. This verifies that the proposed scheme can better serve moving users with less energy consumption by optimizing the trajectory of the UAV.

Fig. 9 compares the performances of different schemes under different user mobility speeds with 10 users. As shown in Fig. 9(a) and (b), higher speed induces more frequent topology variations, degrading the average LoS probability and throughput for all schemes. Nevertheless, PKTD3-TD algorithm consistently outperforms the benchmark methods, benefiting from its continuous control mechanism and smooth trajectory adaptation, and the UAV can dynamically adjust its flight path step-by-step based on instantaneous state feedback, rather than relying on vulnerable long-term trajectory predictions. This step-by-step recalibration endows the system with exceptional fast adaptation capabilities, enabling it to tightly track fast-moving users and maintain reliable communication links even at high speeds. Fig. 9(c) shows that higher mobility increases energy consumption due to frequent maneuvers. Compared with the baselines, PKTD3-TD achieves lower energy consumption by optimizing continuous 3-D trajectories and avoiding inefficient flight maneuvers. Consequently, Fig. 9(d) demonstrates that PKTD3-TD attains the highest normalized weighted difference between throughput and energy consumption across the entire speed range, demonstrating its ability to effectively balance communication performance and energy efficiency in highly dynamic environments.

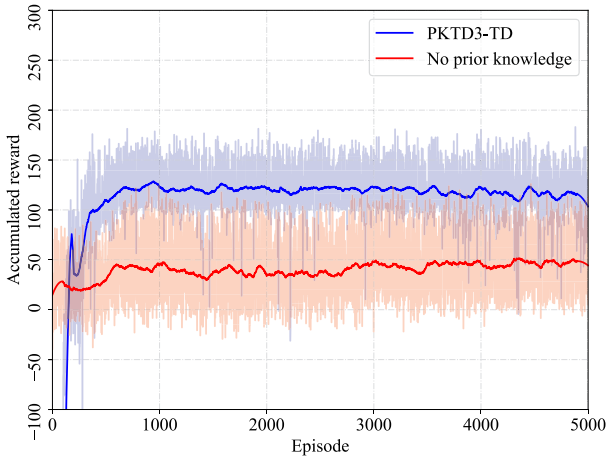
Fig. 10 demonstrates the effectiveness of prior knowledge on algorithm convergence in the cases of different and same starting and destination points. Fig. 10(a) shows that the PKTD3-TD algorithm gradually converges after 2000 episodes when the starting and destination points are different. Incorporating prior knowledge significantly accelerates convergence and improves cumulative rewards by guiding the agent's exploration. Additionally, we compare the cumulative reward curves when $R_{rand} = 10000$, 20000, and 30000, and it is evident that when $R_{rand} = 20000$, the PKTD3-TD algorithm converges fastest. Therefore, we set $R_{rand} = 20000$. Fig. 10(b) shows the scenario with the same starting and ending points. We can observe that the PKTD3-TD algorithm with prior knowledge outperforms the one without prior knowledge in both convergence performance and accumulated reward.

Fig. 11 shows the accumulated reward per episode during the training phase with different numbers of users. It is observed that the accumulated reward increases rapidly for each scenario with varying numbers of users. After approximately 2000 episodes of training, the accumulated reward gradually stabilizes. As depicted in Fig. 11, the proposed PKTD3-TD demonstrates similar convergence performance under different user scenarios. It is adaptable and has good convergence.

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(a) Effectiveness of prior knowledge on algorithm convergence when the starting and ending points are different.



(b) Effectiveness of prior knowledge on algorithm convergence when the starting and ending points are same.

Fig. 10. Effectiveness of prior knowledge on algorithm convergence in different cases. (a) Effectiveness of prior knowledge on algorithm convergence when the starting and ending points are different. (b) Effectiveness of prior knowledge on algorithm convergence when the starting and ending points are same.

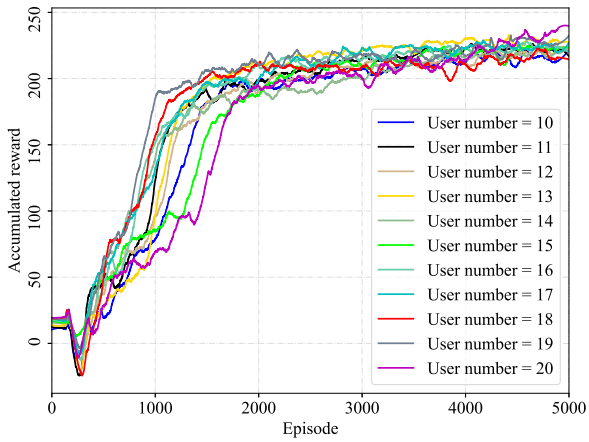


Fig. 11. Accumulated reward versus episode with different numbers of users.

TABLE IV
IMPACT OF DIFFERENT USER POSITION ERRORS ON SYSTEM PERFORMANCES

Error Levels	0	0.01	0.02	0.03	0.04	0.05
Los probability	0.88	0.87	0.87	0.86	0.83	0.83
Throughput ($\times 10^7$ bits)	2.74	2.71	2.71	2.72	2.70	2.70
Energy(J)	133.83	136.59	134.10	138.92	144.78	146.71

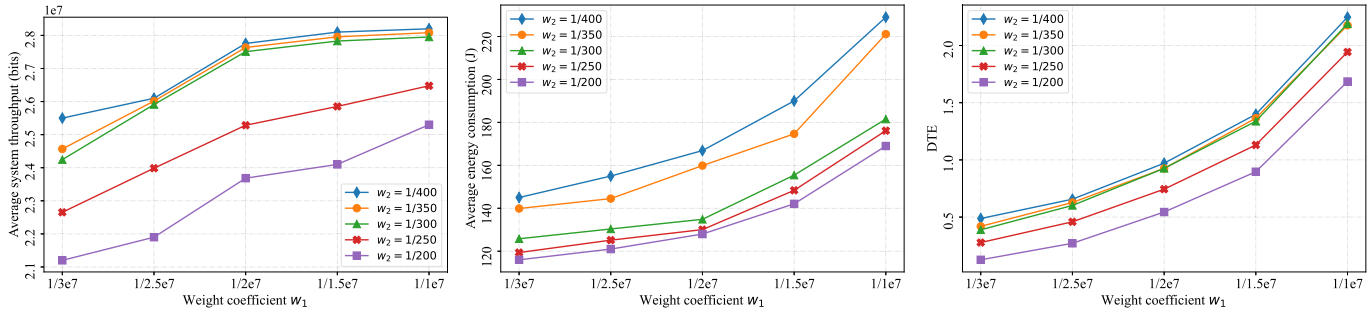
TABLE V
IMPACT OF DIFFERENT SENSING LATENCIES ON SYSTEM PERFORMANCES

Latency Levels	0	1	2	3	4	5
Los probability	0.88	0.89	0.84	0.86	0.83	0.82
Throughput ($\times 10^7$ bits)	2.74	2.76	2.71	2.72	2.69	2.67
Energy(J)	133.83	136.02	139.22	137.59	142.88	148.84

Fig. 12 illustrates the sensitivity analysis of the different weight coefficients w_1 and w_2 on system performances. Fig. 12(a)–(c) demonstrate that under the fixed w_2, increasing w_1 improves throughput by emphasizing communication performance, while also increasing energy consumption due to more aggressive UAV movements. As a result, the DTE increases with w_1 reflecting the enhanced contribution of throughput in the composite objective. Fig. 12(d)–(f) show the impact of w_2 on system performances. It is observed that increasing w_2 effectively reduces energy consumption due to the stronger penalty on UAV movement. However, this also leads to a decrease in throughput, as the UAV becomes more conservative in trajectory adjustment. Consequently, the DTE decreases with increasing w_2, indicating that excessive emphasis on energy saving may degrade overall system performance. Overall, the weight coefficients play a key role in controlling the trade-off between throughput and energy consumption. A larger w_1 favors throughput-oriented strategies, while a larger w_2 leads energy-efficient decision-making. The simulation results display that when $w_1 = 1/(2 \times 10^7)$ and $w_2 = 1/300$, maximum DTE is achieved, enabling satisfactory communication performance without excessive energy consumption.

To further evaluate the robustness and generalization capability of the proposed PKTD3-TD scheme in practical deployment scenarios, we introduce non-ideal factors including position error, sensing latency, and mobility mutation. Specifically, Gaussian noise $\sim \mathcal{N}(0, \sigma_{loc}^2)$ is added to user position observations. As shown in Table IV, the proposed algorithm maintains stable throughput even when σ_{loc} increases to 0.05. Although the energy consumption rises slightly as the UAV performs more adjustments to compensate for the uncertainty, the DRL agent successfully maintains high-quality connectivity. Table V shows the impact of different sensing latencies τ on system performances. The results exhibit slight decreases but remain stable, which is mainly attributed to the trajectory learning based on prior information and the DRL strategy oriented towards long-term

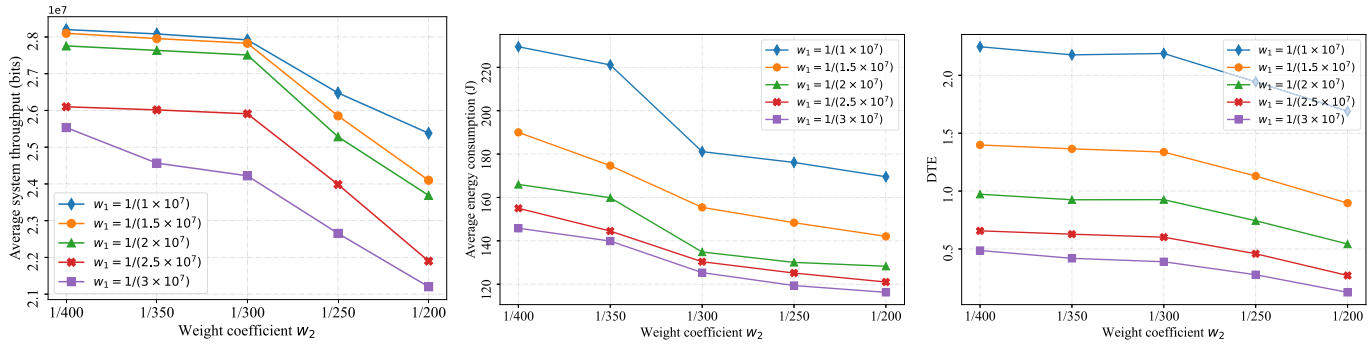
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(a) Average system throughput versus different weight coefficient w_1.

(b) Average energy consumption versus different weight coefficient w_1.

(c) Average DTE versus different weight coefficient w_1.



(d) Average system throughput versus different weight coefficient w_2.

(e) Average energy consumption versus different weight coefficient w_2.

(f) Average DTE versus different weight coefficient w_2.

Fig. 12. Impact of different weight coefficients on system performances. (a) Average system throughput versus different weight coefficient w_1. (b) Average energy consumption versus different weight coefficient w_1. (c) Average DTE versus different weight coefficient w_1. (d) Average system throughput versus different weight coefficient w_2. (e) Average energy consumption versus different weight coefficient w_2. (f) Average DTE versus different weight coefficient w_2.

TABLE VI
IMPACT OF DIFFERENT MOBILITY MUTATIONS ON SYSTEM PERFORMANCES

Mutation Magnitude	**0.2**		**0.6**		**1.0**	
Mutation Probability	**0.02**	**0.06**	**0.02**	**0.06**	**0.02**	**0.06**
Los probability	0.86	0.82	0.85	0.80	0.82	0.78
Throughput ($\times 10^7$ bits)	2.74	2.70	2.73	2.69	2.71	2.66
Energy(J)	131.39	135.51	133.92	136.51	137.85	146.15

rewards. This strategy does not rely on the instantaneous users' positions but utilizes long-term feedback to make decisions for trajectory, thereby reducing sensitivity to short-term observation errors and delays of user positions. To assess the generalization in highly dynamic environments, Table VI illustrates the performances under sudden user maneuver scenarios, characterized by the mutation magnitude and probability. The results show that the proposed scheme adapts rapidly to these unpredicted changes, with only minor fluctuations in throughput and energy consumption. This is enabled by the combination of prior knowledge and stochastic exploration, effectively allowing the UAV to continuously absorb fresh environmental interactions and adjust its trajectory to overcome sudden changes. This proves that the learned policy possesses excellent robustness and generalization for complex and dynamic scenarios.

VI. CONCLUSION

In this paper, we investigate 3-D trajectory design for a UAV-assisted communication network with randomly moving ground users. Due to the uncertainty of user movement, we consider a variety of practical influencing factors and formulate the optimization problem with the goal of maximizing the network throughput while minimizing UAV energy consumption. Due to the high-dimensional variables in 3-D space and user mobility, the optimization problem is difficult to be solved. So, we reformulate it as an MDP, then adapt DRL methods and design the PKTD3-TD algorithm based on the TD3 framework to achieve optimal solutions. Specifically, we introduce prior knowledge into the algorithm and design prior policies for UAV learning to improve the convergence speed and algorithm robustness. Numerical results demonstrate the effectiveness of the proposed scheme in terms of network throughput and UAV energy consumption. Despite the effectiveness of the proposed PKTD3-TD framework, several limitations remain. As a DRL-based approach, the UAV control policy is learned during an offline training phase, which requires a certain amount of interaction data and computational resources. When the operating environment undergoes long-term or significant changes, the policy learned from the original training distribution may struggle to consistently maintain near-optimal performance. In the future work, we will further explore more adaptive mechanisms that leverage previously learned experiences to enhance policy

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adaptability. Without altering the original network architecture, a limited amount of interaction data collected in new scenarios can be utilized to selectively adjust the mapping from system states to continuous motion decisions, thereby enhancing the adaptability in dynamic environments for UAV-assisted communication networks.

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